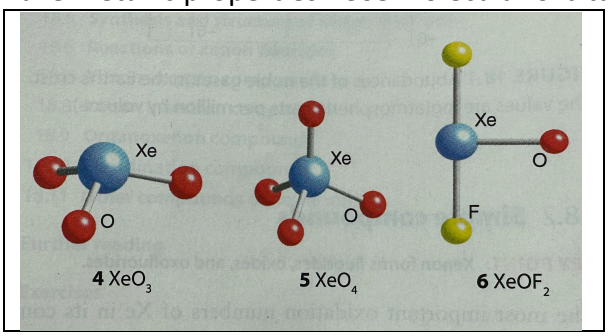
Inorganic Cumulative Exam
April 7 2023

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

1. (a) (10 pts) Explain how metallic bonding is different than ionic or covalent bonding?
   (b) (10 pts) Lithium (solid) is metallic while the diatomic molecule, Li_2 (gas), does not have metallic properties. Use molecular orbital theory to explain this.



2. (a) (4 pts each) Assign a point group to each of the molecules (XeO_3, XeO_4, $XeOF_2$) shown to the left.
   (b) (8 pts) What is meant by a "high symmetry" point group? Give three examples of high symmetry point groups.

3. (a) (12 pts) Use molecular orbital theory to derive the molecular orbital diagram of H_2S, including annotations for which atomic orbitals contribute to each molecular orbital. For this problem, you need to know that the orbital energies for the elements are -13.6 eV (H 1s), -22.71 eV (S 3s), and -11.62 eV (S 3p).
   (b) (4 pts each) Sketch the electron densities corresponding to the LUMO and HOMO levels of H_2S. indicating the nuclei and contributing atomic orbitals.

4. Sketch all of the possible isomers for an "octahedral" molecule with the formula of MA_3B_3 (M = central metal; B,C = monatomic ligands) and assign the point group to each.
5. Assume that the hypothetical molecule, H_6, has the same hexagonal structure as benzene.
   (a) (12 pts) Draw the linear combinations of 1s orbitals for the lowest-energy and highest-energy molecular orbitals. Explain your rationale using quantum mechanical arguments.
   (b) (8 pts) Would this hexagonal H_6 molecule be more or less stable than a linear H_6 molecule? Explain your rationale for this using quantum mechanical arguments.

6. Metallic Li (s) forms a body-centered cubic lattice with a lattice constant of 330 pm. Using these observations, calculate the density of metallic Li (s) (in units of g cm^{-3})?
7. Do you expect fewer defects to form in a solid crystal near the crystal's melting point? Please explain your answer in detail.
8. (a) (8 pts) If the conductivity of room-temperature Cu (s) is $\sigma = 5.9 \times 10^7$ S m^{-1}, what is the resistivity of Cu, ρ?

(b) (12 pts) What is the resistivity (in Ohms) of a Cu wire with 3 mm diameter and 1 m length? (If you do not know how to solve (a), assume some value for ρ to answer this question.)

9. What is electroluminescence? In your answer, provide two examples of compounds that are used in electroluminescence applications.
10. (a) (10 pts) Band gaps in semiconductors can either be *direct* or *indirect*. What is meant by this distinction?
    (b) (10 pts) How does the direct/indirect nature of a semiconductor affect how it would be used in a photodetector or photovoltaic device?

PERIODIC TABLE OF ELEMENTS

Atomic # / Symbol / Name / Weight

- C Solid
- Hg Liquid
- H Gas
- Rf Unknown

Metals: Alkali metals; Alkaline earth metals; Lanthanoids (Lanthanides); Actinoids (Actinides); Transition metals; Post-transition metals

Metalloids

Nonmetals: Other nonmetals; Noble gases

Pnictogens; Chalcogens; Halogens

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H Hydrogen 1.008																	2 He Helium 4.0026
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122											5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305											13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
7	87 Fr Francium (223)	88 Ra Radium (226)	89–103	104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

6	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97
7	89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)

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